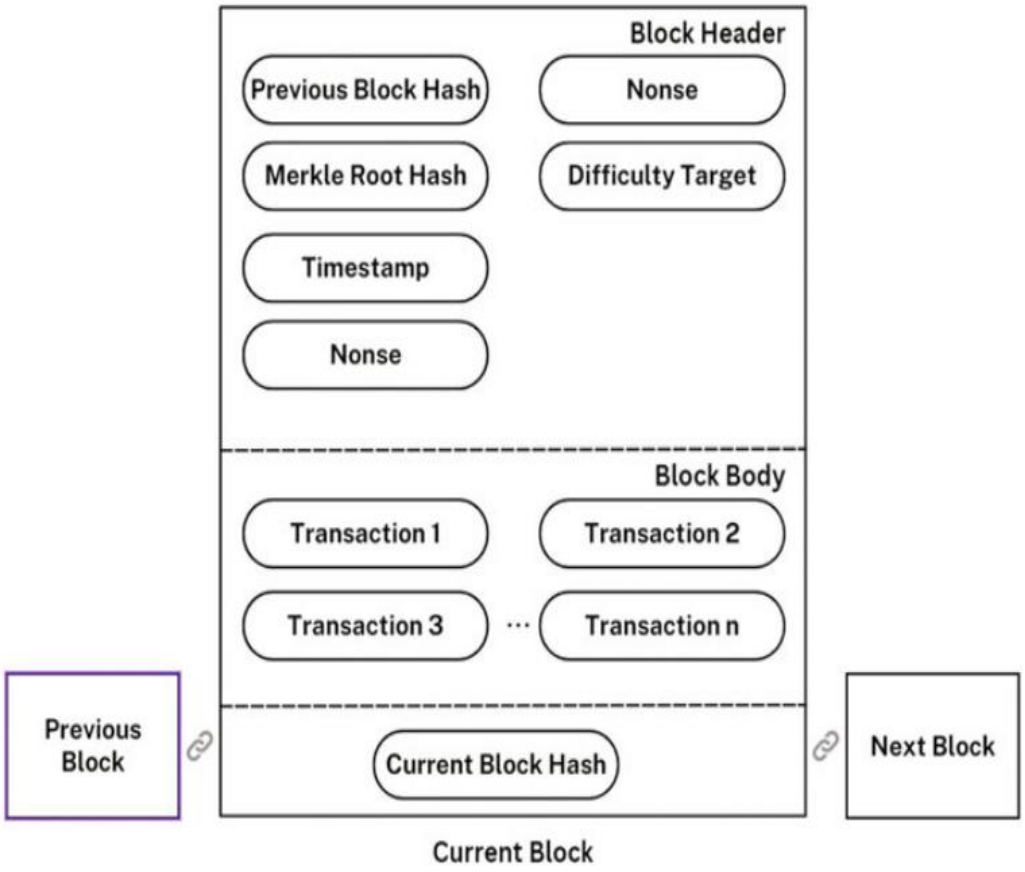
Block and Block Structure

Block

- A block is an essential component of a blockchain, typically comprising a collection of transactions.
- Its structure consists of a preamble that stores metadata such as a timestamp and reference to the previous block (known as the parent block).
- and a body that contains the actual transaction data.
- The unique cryptographic hash of each block ensures the chain's integrity.
- Blocks are chained together in chronological order to create a secure and immutable ledger.
- Once the data within a block are validated, the block is considered closed, and a new block is created to accommodate and validate new transactions.

Block Structure

A block typically consists of several components as shown in the figure. The structure of a block in a blockchain can vary depending on the specific implementation and type of blockchain.



Block Structure

Block header: The block header contains metadata about the block, including the block number, timestamp, and the hash of the previous block in the chain.

Nonce: A nonce is a random number generated by miners in order to solve the cryptographic puzzle required to add a block to a blockchain.

Transaction data: The transaction data section contains the actual data that are being added to the blockchain. This can include information such as the sender and recipient addresses, the amount of cryptocurrency being transferred, and any additional data related to the transaction.

Block Structure

Block hash: The block hash is a unique identifier that represents the contents of the block. It is generated by hashing the block header and transaction data using a specific cryptographic algorithm.

Merkle tree: The Merkle tree is a data structure used to efficiently store and verify large amounts of transaction data. The transactions are hashed and combined in pairs to form a series of hashes, which are then combined until a single root hash is produced.

Ledger

- A ledger is a digital record-keeping system that records all transactions made on the network.
- The ledger maintains a permanent and tamper-proof record of all transactions.
- They are often referred to as distributed ledgers.
- The ledger is decentralized; no central authority controls it.
- All nodes on the network have a copy of the ledger. The copy is updated in real time as long as new transactions are made.
- All nodes can view the transactions and their associated data.
- Ledger in a blockchain is immutable.

Shared Distributed Ledger

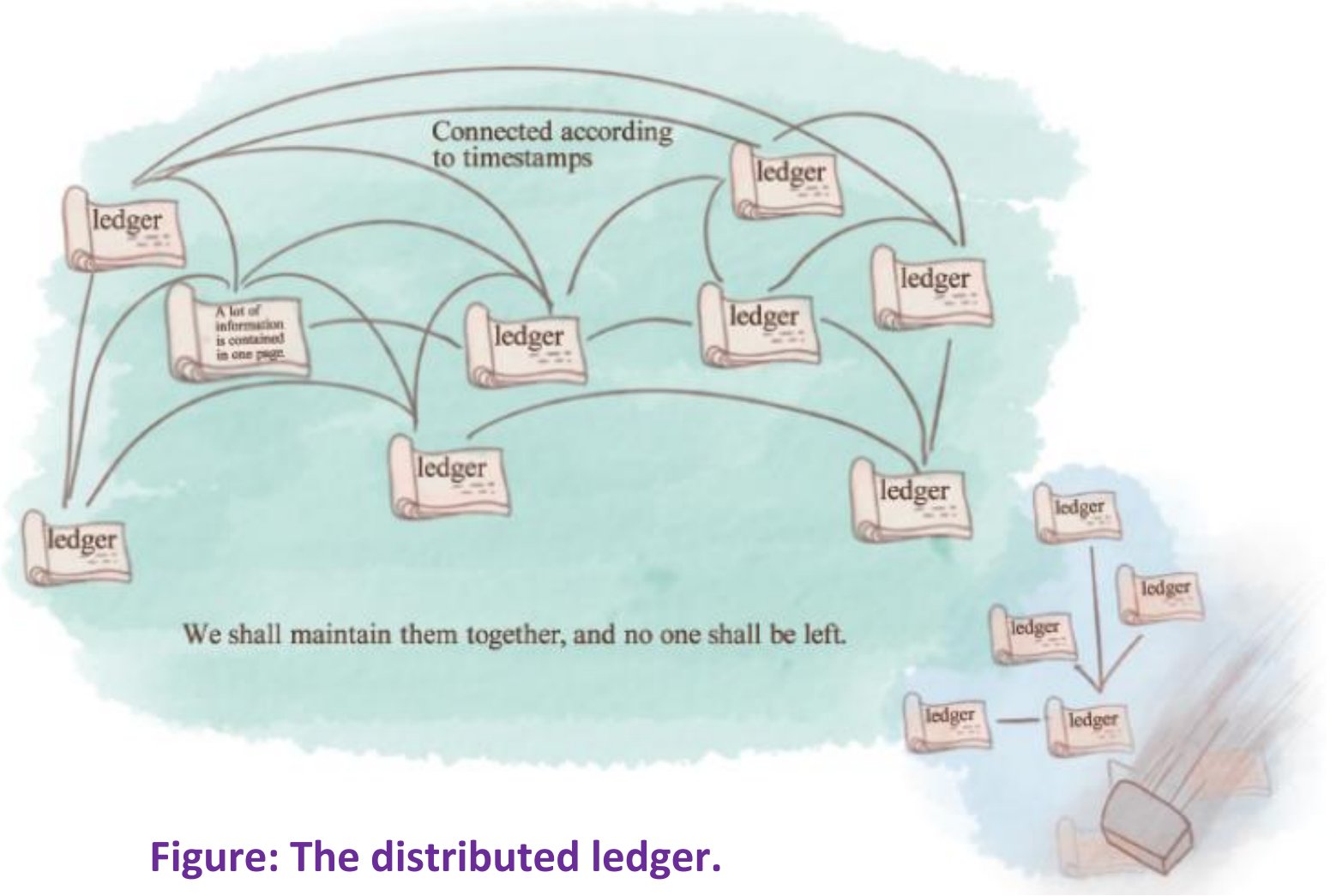


Figure: The distributed ledger.

Distributed

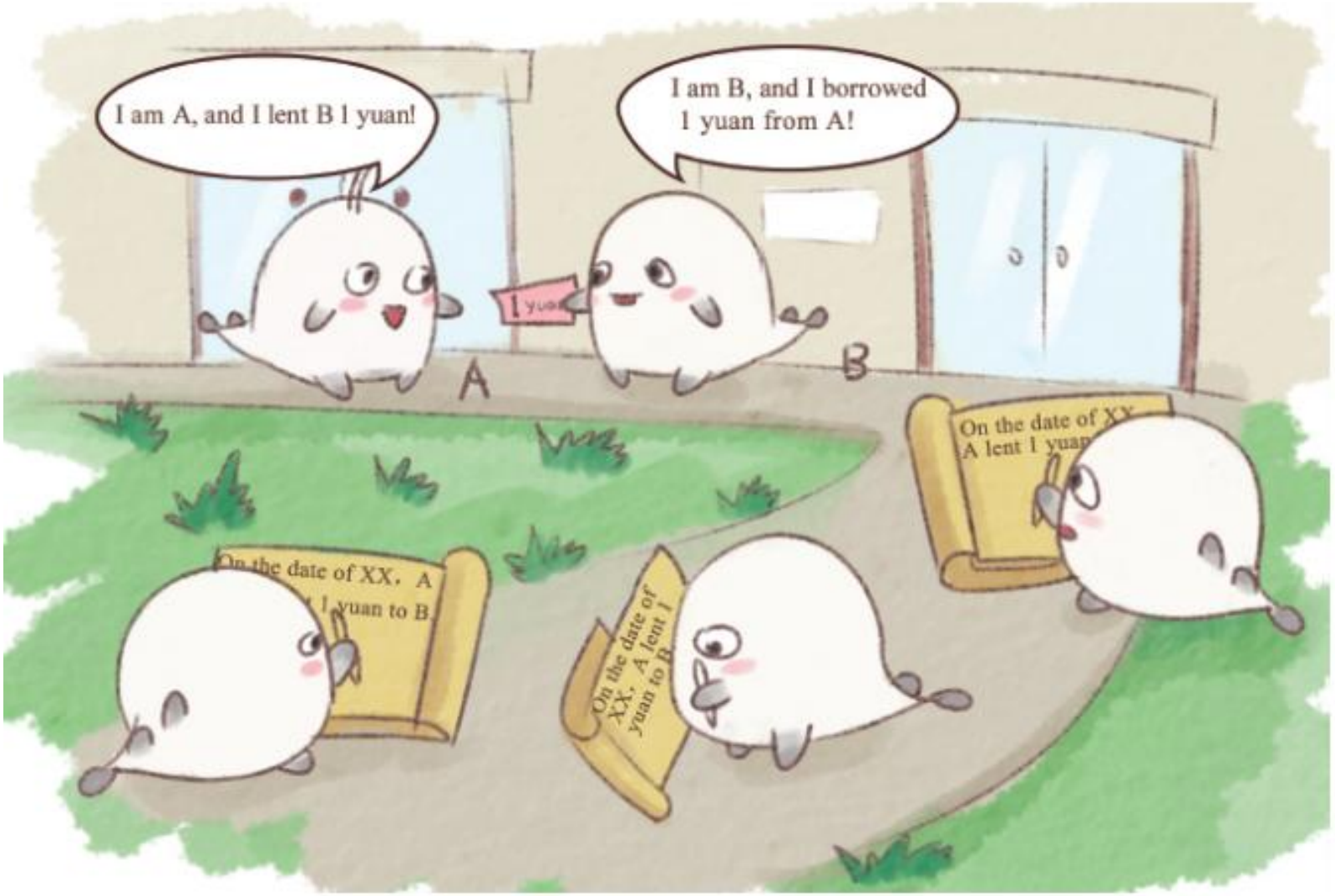
- Distributed refers to the way in which data are stored and processed across a network of computers or nodes.
- Rather than being centralized in a single location,the workload is shared across multiple nodes.
- Each node has a copy of the entire blockchain, and each node is responsible for verifying transactions and maintaining the integrity of the network.

How distributed Ledger Created?

Assume that we have a decentralized city with five residents—we name them A, B, C, D, and E. Here is what they do when they want to borrow money from others:

Suppose resident B borrows one yuan from resident A, and A informs other residents by shouting, “I am A, and I just lent one yuan to B.” Subsequently, B will say loudly, “I am B, I just borrowed one yuan from A.” When other residents—C, D, and E—hear what they say, they write down on their ledgers, “dd/mm/yy, A lends one yuan to B.”

How distributed Ledger Created?



Benefits of Distributed Architecture

Resource sharing: Computers connect and share resources like software and hardware. Resource sharing involves remote access to components, reducing costs and improving convenience. Data sharing ensures consistency and facilitates information exchange.

Heterogeneity: Varying hardware, programming languages, networks, and implementations, all working together efficiently.

Scalability: Scalability Accommodates more users and computers, without altering components but designing them to handle growth effectively.

Concurrency: Enables simultaneous execution of multiple activities across different machines, managed by a common system, allowing for parallel processing.

Fault tolerance: Distributed systems enhance fault tolerance and availability through software recovery and hardware redundancy, ensuring reliable operation despite failures.

Openness: Openness in distributed systems refers to their capability to adapt and enhance hardware and software components as needed, enabling seamless integration of new components through standardized interfaces. This ensures compatibility and allows for resource sharing services.

Transparency

In a blockchain network, transactions are public and visible to all nodes on the network. This means that anyone can view the transaction data, including the sender, receiver, amount, and all other relevant details. Transparency provides the benefits of---

Trust: The transparency of blockchain instills trust by allowing anyone to verify the validity and proper execution of transactions.

Accountability: The public nature of blockchain ensures accountability for all participants' actions on the network.

Fraud prevention: The transparency of blockchain makes it more difficult for bad actors to commit fraud or engage in other malicious activities on a network.

Efficiency: The transparency of blockchain improves network efficiency by eliminating intermediaries and reducing transaction costs.

Confirmation

- Block confirmation refers to the process of validating and adding transactions to a blockchain.
- It involves solving mathematical puzzles, ensuring transaction integrity, and preventing double-spending.
- Validated transactions are added to a block, which is then appended to the blockchain through the process of mining.
- Block confirmation is crucial for maintaining the integrity and security of the blockchain. Once a transaction is confirmed and added to the blockchain, it becomes permanent and cannot be altered.
- This process eliminates the need for intermediaries, enhances network efficiency, and reduces transaction costs.